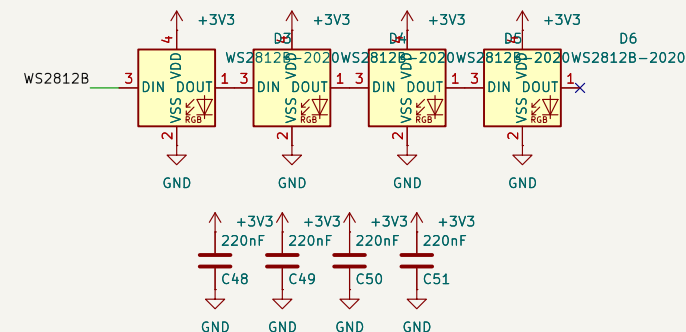
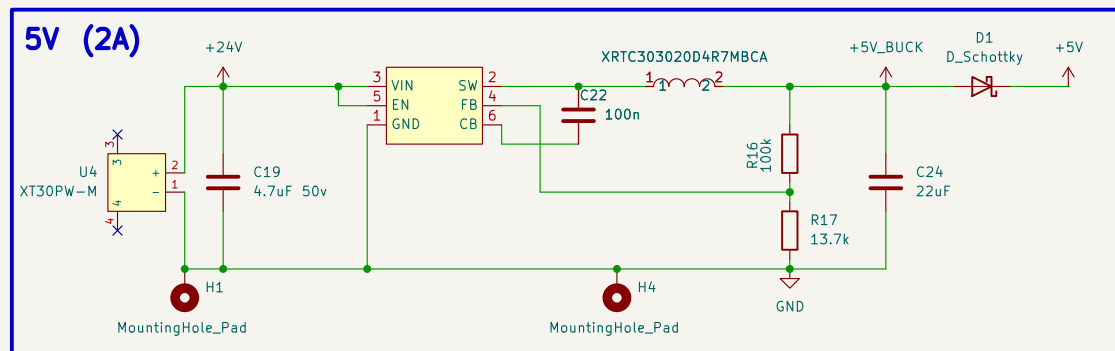
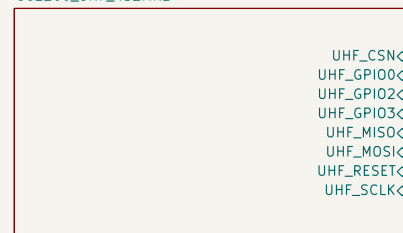


FPWM version of the LMR51420YFDDCR for less output ripple at low voltage and more consistency but less efficient

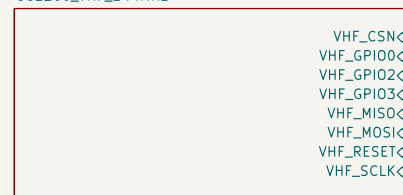


CC1200_UHF_432MHz



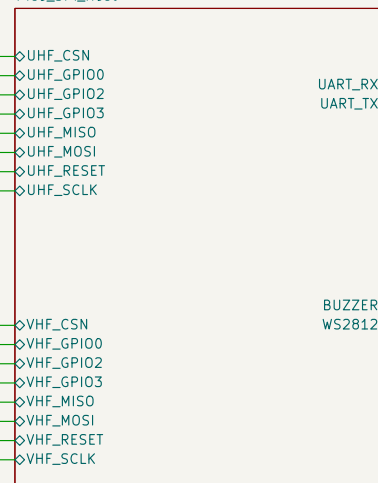
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CC1200_VHF_144MHz



File: CC1200_VHF_144MHz.kicad_sch

Pico_SPI_Host



File: Pico_SPI_Host.kicad_sch

Pi_Header



File: Pi_Header.kicad_sch

UART_RX
UART_TX

BUZZER
WS2812B

+3V3
D2
1N4148WS

+3V3
Q1
BSS138
GND

J4
Conn_01x01

J5
Conn_01x01

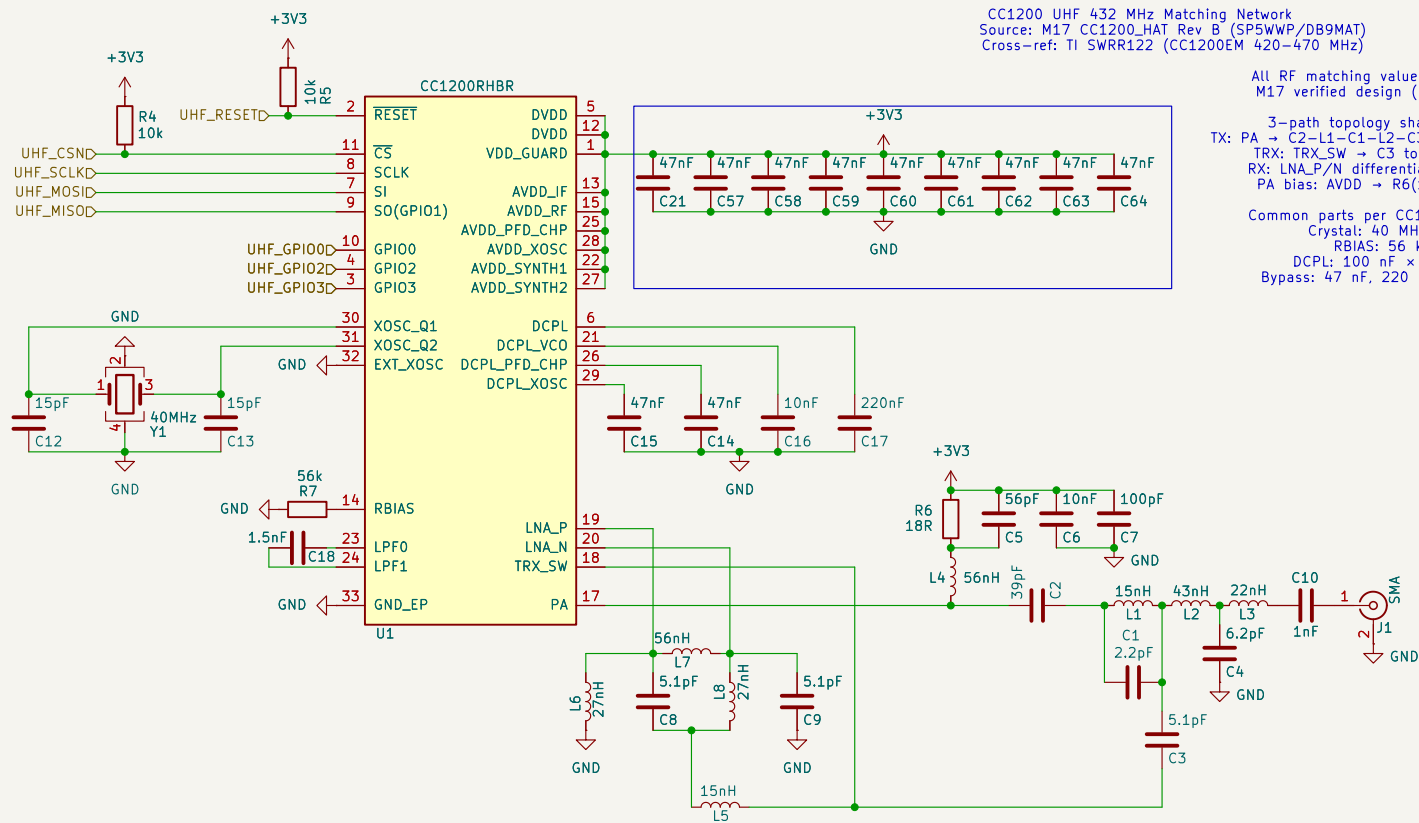
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Title: RF_HAT

Size: A4 Date:

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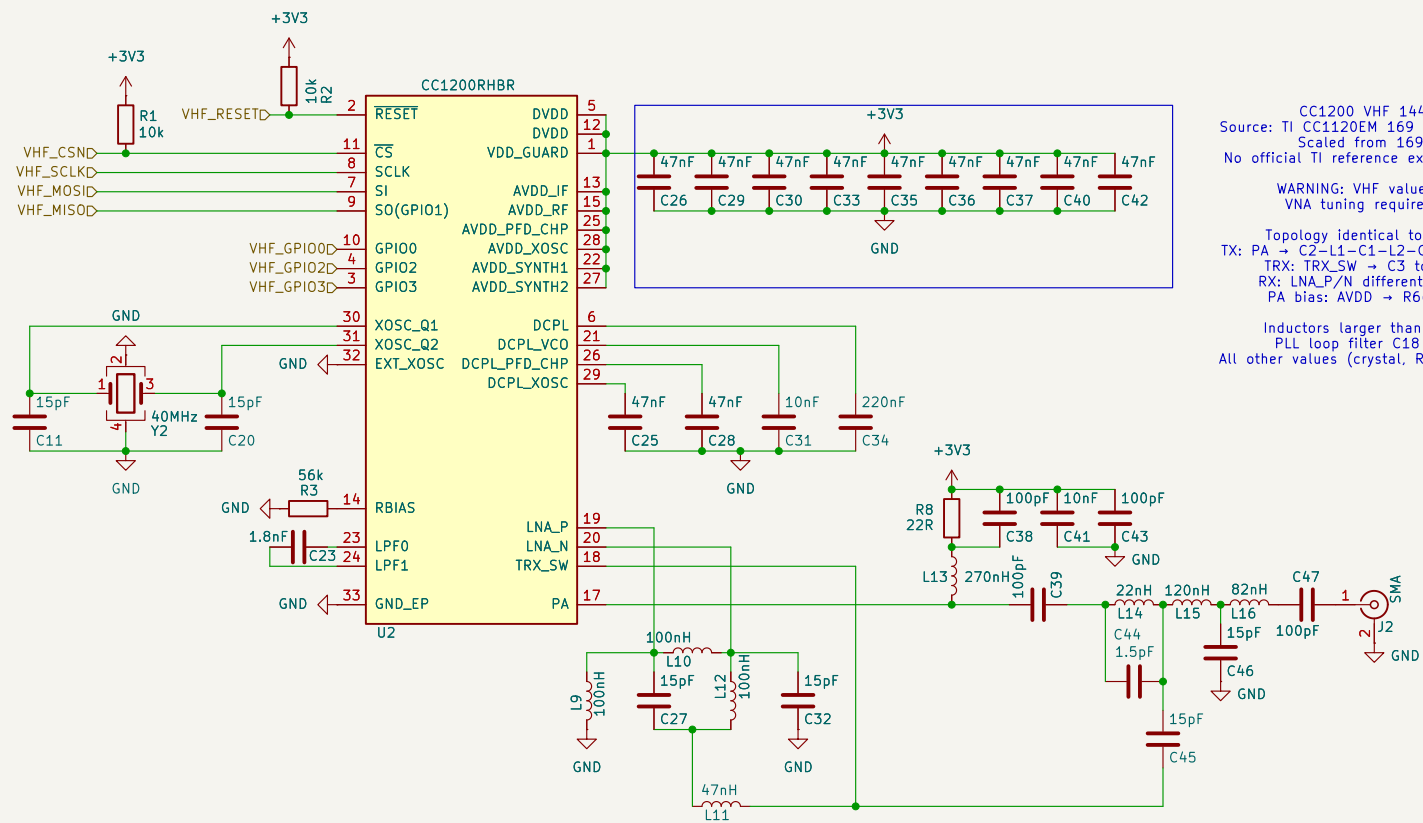
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File: CC1200_UHF_432MHz.kicad_sch

Title: RF_HAT

Size: A4
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Id: 2/5



CC1200 VHF 144 MHz Matching Network
Source: TI CC1120EM 169 MHz ref design (TIDR220/TIDR222)
Scaled from 169 MHz to 144 MHz band.
No official TI reference exists for CC1200 at 136-160 MHz.

WARNING: VHF values are starting points only.
VNA tuning required during board bring-up.

Topology identical to UHF (3-path, shared SMA):
TX: PA → C2-L1-C1-L2-C3-L3-C4 → C10(DC block) → SMA
RX: TRX_SW → C3 to TX path, L5 to balun center
RX: LNA_P/N differential balun (L6, L7, L8, C8, C9)
PA bias: AVDD → R6(22Ω) → L4 → PA, C5 bypass

Inductors larger than UHF (lower freq = more L).
PLL loop filter C18 = 1.8 nF (vs 1.5 nF UHF).
All other values (crystal, RBIAS, DCPL, bypass) same as UHF.

Sheet: /CC1200_VHF_144MHz/
File: CC1200_VHF_144MHz.kicad_sch

Title: RF_HAT

Size: A4
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